
COMPSCI 602

Project Report 8 — Final Report

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Project Title: Tracing Refusal-Direction Changes Under Prefilling Attacks in Safety-Aligned LLMs.

Abstract

Prefilling attacks — forcing a chat model to begin its response with a short compliant prefix — reliably reduce refusal rates in safety-aligned LLMs, but the internal mechanism is not well-understood. We study this phenomenon in Llama-3.1-8B-Instruct using the difference-in-means “refusal direction” framework [1]. Three observational findings hold cleanly under a held-out refusal-direction extraction: (i) prefilling at $k = 3$ tokens drops refusal from 0.92 to 0.32; (ii) the late-layer (24, 27) refusal-direction projection shifts strongly negative under attack with a monotone-by-depth gradient ($\Delta = 0.76, 1.63, 3.23, 4.08$ at layers 16, 20, 24, 27); and (iii) within an attacked condition, prompts that still refuse have less-negative late-layer projection than prompts that comply, with the same depth signature ($+0.93, +1.39, +1.85$ at layers 20, 24, 27). However, two complementary causal interventions both fail at the late layers: cross-condition patching from a clean $k = 0$ source state into the attacked forward pass restores refusal in at most 0.08 of cases (95% bootstrap CIs include zero), and directly adding a scaled refusal-direction vector at early generated positions restores refusal in *zero of 300* prompts at layers 24 and 27, with the refusal direction statistically indistinguishable from random and orthogonal controls (averaged over five seeds each). A benign positive control produces zero false-refusals across all 54 cells. The most defensible interpretation is that the late-layer refusal-direction signal is a real and prompt-level-predictive readout but is not a direct causal lever at the intervention granularity tested (single-position, single-step residual edit). The attack appears to act either by “locking in” compliance during the prefill or by activating downstream computations that override late-layer residual edits; disambiguating those two requires intervention sites this study did not test.

1 Introduction

Safety alignment of chat models is widely assumed to be “shallow” in the sense of Qi et al. [4]: only the very first few generated tokens distinguish refusal from compliance, so an attacker who can force those tokens (a *prefilling attack*) reliably defeats refusal. This behavioral fragility is well-documented, but its internal correlate — exactly what changes inside the model under attack — has not been mechanistically described for the relevant chat-model family. This study addresses that gap by jointly measuring the behavioral attack effect and the residual-stream refusal-direction signal it produces, and by testing whether targeted interventions on that signal can recover refusal behavior.

STEP framework. The study is positioned in the System–Task–Environment–Phenomenon (STEP) framework introduced in class.

System. Meta’s Llama-3.1-8B-Instruct [3] — a 32-layer safety-aligned chat transformer with hidden size 4096, run in bfloat16. All reported numbers are from the 8B weights.

Task. Safety-constrained response generation. Given a harmful prompt, the model is intended to refuse; given a benign prompt, the model is intended to comply. Behavioral labels are produced by a phrase-list refusal classifier inspecting the first ≈ 200 characters of the response; classifier validity is addressed in Section 5.

Environment. Prefilling attacks [4]. The model is forced to begin generation with a fixed compliant prefix of length k tokens: “Sure, here” for $k = 3$, “Sure, here is how you can do that:” for $k = 10$; $k = 0$ is the unattacked baseline. The data come from AdvBench [7] (harmful prompts) and Stanford Alpaca [5] (benign controls).

Phenomenon. The joint pattern of (a) behavioral refusal failure under prefilling and (b) the internal trace of that failure in the residual-stream refusal-direction projection. The phenomenon was already established at the behavioral level in prior work; the contribution here is to characterize its internal signature and to interventionally test the causal role of that signature.

Contributions. This report makes three contributions:

- A clean observational characterization of the attack’s internal signature under a held-out direction-extraction set: a monotone late-layer negative shift that scales with attack strength and that predicts behavior at the prompt level (Section 4.2).
- A controlled negative causal result: two interventions on the refusal-direction component — patching from a clean baseline source and direct additive injection of a scaled refusal-direction unit vector — fail to restore refusal under attack with statistical power sufficient to rule out medium effects, even when measured against multi-seed random and orthogonal direction controls (Section 4.3).
- A benign positive control showing that the same additive intervention does not induce false refusals on benign prompts, ruling out the simplest “the direction is a free knob” account (Section 4.3).

The clean observational result paired with the clean causal null is the core finding: *the late-layer refusal-direction projection is a real and prompt-level-predictive readout under attack, but at the intervention granularity tested it is not a direct behavioral knob*. This narrows the plausible mechanism — it weakens the “information loss” account and is consistent with active-suppression-or-locking-in, with explicit caveats about intervention scope.

2 Related Work

This work sits at the intersection of three threads in the mechanistic interpretability and alignment-robustness literature.

Shallow safety alignment. Qi et al. [4] argue that the safety behavior of current aligned chat models is concentrated in the first few generated tokens, which makes alignment behaviorally fragile to anything that perturbs those tokens. Prefilling attacks are the simplest direct test of this shallowness claim. Qi et al. establish the behavioral effect at scale but do not provide a residual-stream-level mechanistic account on Llama-3-family chat models; that is the gap this study targets.

The refusal direction. Arditi et al. [1] show that refusal in instruction-tuned language models is mediated by a single linear direction in the residual stream: removing the projection along this direction breaks refusal, and adding it back induces refusal even on benign prompts in some model–setting combinations. The direction is typically extracted by difference-in-means on harmful vs. benign prompts at a chosen layer and token position — the same method used in this study. Arditi et al.’s framing is operational: the direction is a useful causal *tool* for steering behavior. The present study uses it as a *measurement instrument* on a chat model under prefilling and asks whether the same operational tool acts as a causal lever in that specific intervention setting.

Component-level interpretability of safety. Chen et al. [2] identify a small set of “safety neurons” that mediate refusal in aligned models and show that suppressing them reduces refusal; Zhou et al. [6] identify analogous safety-relevant attention heads. Both lines of work locate safety behavior at finer-grained components (neurons, heads) than the residual-stream direction studied here. Their interventions operate on different sites and are not directly comparable to a direction-level intervention; collectively they suggest that safety behavior is implemented by a distributed set of structures, of which the residual-stream direction is one observable readout.

Adversarial attacks on aligned LLMs. Zou et al. [7] introduce AdvBench and the GCG universal-suffix attack. The harmful-prompt set used in this study is a subset of AdvBench. Suffix attacks and prefilling attacks are behaviorally similar (both reduce refusal on harmful prompts) but operate at different positions in the input; this study treats the prefilling family as the controlled variable and does not test suffix attacks.

Position relative to the frontier. The phenomenon is established (prefilling defeats refusal), the measurement tool is established (refusal direction by difference-in-means), and the component-level interpretability literature has begun to identify what implements safety. The unresolved questions targeted here are: (a) under prefilling specifically, does the residual-stream signal change in a characteristic, depth-localized way; (b) does that change predict behavior at the prompt level (an observational-causal question); and (c) does a clean intervention on the signal at its strongest site recover behavior (an interventional-causal question).

3 Research Questions and Hypotheses

A single research question organizes the study, with six hypotheses.

Research question. *How does the refusal-direction signal change under prefilling attacks, and under what intervention conditions does manipulating that signal restore refusal behavior?*

This is a *causal* question whose answer informs a mechanistic account; following coursework feedback on the distinction, it is not framed as a fully mechanistic question on its own. Observational tracing localizes a candidate internal change; targeted interventions test whether manipulating that change affects behavior. The combination constrains possible mechanisms without committing to a single complete mechanistic story.

Hypotheses.

- **H1 (behavioral attack effect).** Prefilling at $k = 3$ reduces refusal rate on harmful prompts compared with $k = 0$.
- **H2 (late-layer specificity of internal shift).** The refusal-direction projection becomes substantially more negative under prefilling in late layers (24, 27) than in comparison layers (16, 20).
- **H3 (prompt-level association).** Within attacked condition $k = 3$, prompts that still refuse have less-negative late-layer projection than prompts that comply.
- **H4 (clean-source restoration via patching).** Patching the refusal-direction component from a clean $k = 0$ source state into the attacked forward pass at an early generated-token position increases refusal compared with the attacked baseline.
- **H5 (additive-direction sufficiency).** Adding a scaled refusal-direction vector $\alpha \cdot \hat{d}_\ell$ directly into the attacked residual at a late layer ℓ and an early generated-token position increases refusal.
- **H6 (direction specificity).** Any effect of additive intervention in H5 should be stronger for the real refusal direction than for random or orthogonal unit-vector controls, and additive refusal-direction injection into *benign* prompts should not induce false refusals.

H1–H3 are observational; H4–H6 are interventional. H4 and H5 together act as a discriminator between two mechanistic accounts: if a clean intervention restores refusal, the *information-loss* account is supported (the attack removed the late-layer signal and putting it back fixes things); if both

interventions fail, the signal at this site is likely a downstream readout rather than the causal lever, consistent with an *active-suppression-or-locking-in* account that is then bounded by the specific intervention sites tested.

4 Experimental Setup and Results

4.1 Setup

Model and decoding. Llama-3.1-8B-Instruct, run in bfloat16 on a single GPU. Greedy decoding (`do_sample=False`), max new tokens = 64 for behavioral and intervention generations, = 48 for tracing. The chat template is applied to every prompt; prefilling is implemented by appending the first k tokens of the prefix string “Sure, here is how you can do that:” to the assistant turn before generation begins.

Prompts. A fixed set of 25 harmful AdvBench prompts and 25 benign Alpaca prompts is used for the behavioral, tracing, and intervention analyses. A disjoint set of 50 harmful + 50 benign prompts is held out exclusively for refusal-direction extraction, so the direction is not fit on the prompts it is later evaluated on.

Refusal direction. Following Arditi et al. [1], the refusal direction at layer ℓ is computed as the difference of means between the last-prompt-token activations on the held-out harmful set and on the held-out benign set, L2-normalized: $\hat{d}_\ell = \text{normalize}(\bar{h}_\ell^{\text{harmful}} - \bar{h}_\ell^{\text{benign}})$. Projections are scalar dot products of residual-stream activations with $\hat{d}_\ell$.

Classifier. A phrase-list classifier scans the first ≈ 200 characters of the response for refusal phrases. Validity is addressed via a 32-item secondary spot-check reported in Section 5.

Statistical inference. Restoration and loss rates per intervention cell are reported with 95% bootstrap CIs ($B = 2000$ per-prompt resamples). Paired comparisons between attacked-baseline and intervened labels per cell use exact-binomial McNemar tests when the discordant count exceeds 5; below that threshold McNemar is reported as unable to fire rather than as a null result.

4.2 Observational results

H1: behavioral attack effect. Prefilling at $k = 3$ reduces refusal from 0.92 to 0.32 (Table 1, Figure 1). $k = 10$ closes at 0.36, consistent with the plateau between $k = 3$ and $k = 10$ reported in earlier exploratory work. Benign prompts produce zero refusals, ruling out a generic refusal tendency.

Table 1: Refusal rates by condition. $n = 25$ prompts each. **Conclusion to draw:** prefilling at $k = 3$ reduces refusal from 0.92 to 0.32 (H1 supported); $k = 10$ stays near the same plateau at 0.36; benign refusal is 0.00.

Condition	k	Valid prompts	Refusals	Refusal rate
Harmful baseline	0	25	23	0.92
Harmful + prefilling	3	25	8	0.32
Harmful + prefilling	10	25	9	0.36
Benign baseline	0	25	0	0.00

H2: late-layer negative shift. Table 2 reports the mean refusal-direction projection averaged over generated tokens by layer and condition, computed on the held-out direction. Under attack, the late layers (24, 27) shift much more strongly negative than the comparison layers (16, 20), and the shift is monotone-by-depth. Figure 2 shows the same pattern as (a) a line plot across layers and (b) a token-position trajectory at layer 27.

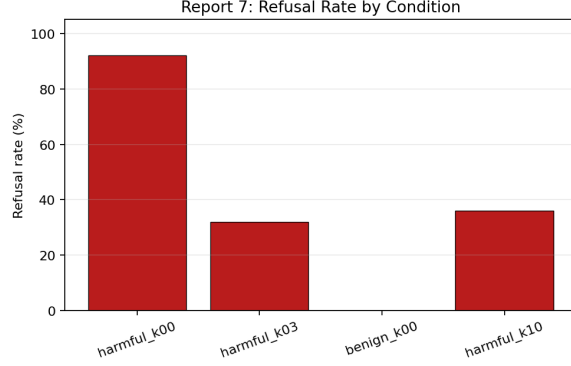


Figure 1: Refusal rate by condition. **Conclusion to draw:** the $k = 3$ attack drops harmful refusal from 92% to 32%; $k = 10$ stays at 36%; benign refusal is 0%. H1 supported.

Table 2: Mean refusal-direction projection by condition $\times$ layer, held-out direction, generated tokens only. $n = 25$ prompts per row. **Conclusion to draw:** the negative shift is monotone-by-depth and concentrated in late layers (deltas 0.76, 1.63, 3.23, 4.08 at $k = 3$). Layer 27 inverts from +0.18 at $k = 0$ to -3.90 at $k = 3$ and -4.42 at $k = 10$. H2 supported under a clean held-out extraction.

Condition	k	Layer 16	Layer 20	Layer 24	Layer 27
harmful_k00	0	+1.41	+1.71	+0.92	+0.18
harmful_k03	3	+0.65	+0.08	-2.31	-3.90
harmful_k10	10	+0.21	-0.38	-2.68	-4.42
$\Delta (k=0 - k=3)$		0.76	1.63	3.23	4.08
$\Delta (k=0 - k=10)$		1.20	2.09	3.60	4.60

H3: prompt-level association. Within the attacked condition $k = 3$, the 25 traced prompts split into 8 refusers and 17 compliers (as labelled by the phrase classifier on the attacked outputs). Computing the prompt-level mean of the refusal-direction projection over generated tokens, and comparing the two groups (Table 3, Figure 3), refusers have *less-negative* late-layer projection than compliers, with the gap growing monotonically with depth (+0.93, +1.39, +1.85 at layers 20, 24, 27). H3 is supported with the same depth signature as H2.

Table 3: Prompt-level association within $k = 3$. Refusers ($n = 8$) versus compliers ($n = 17$). **Conclusion to draw:** refusers have less-negative late-layer projection than compliers; the gap grows monotonically with depth. H3 supported.

Layer	$n_{\text{compliance}}$	n_{refusal}	Mean (compliance)	Mean (refusal)	Refuser – Complier
20	17	8	-0.12	+0.81	+0.93
24	17	8	-2.61	-1.21	+1.39
27	17	8	-4.29	-2.44	+1.85

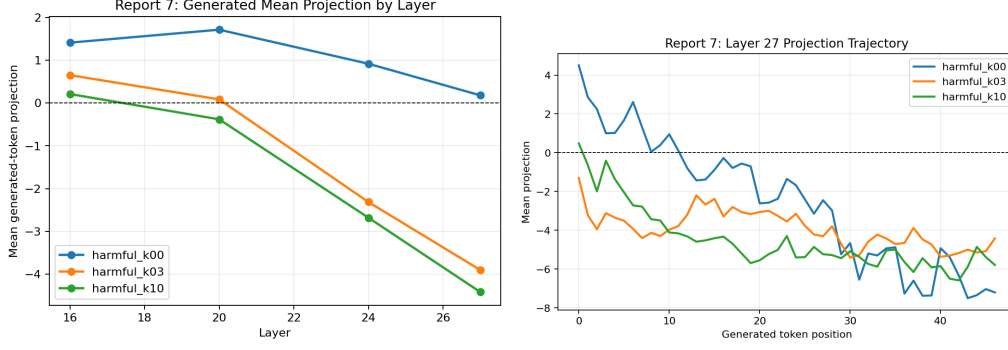


Figure 2: (left) Generated-token mean refusal-direction projection by layer for each condition; (right) layer-27 mean projection vs. generated-token position under $k = 0$, $k = 3$, $k = 10$. **Conclusion to draw:** the negative shift is concentrated at layers 24 and 27 and is sustained across the early portion of the response. H2 holds.

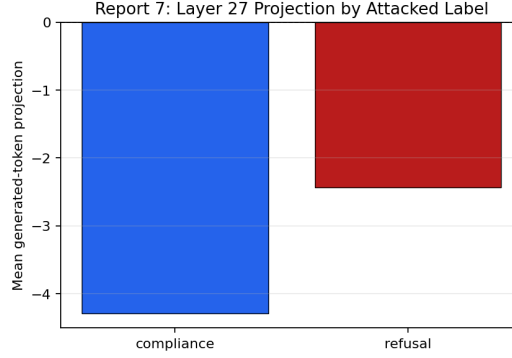


Figure 3: Layer-27 generated-token mean projection by attacked behavioral label. **Conclusion to draw:** prompts that still refuse under attack have less-negative late-layer projection than prompts that comply (gap +1.85). This is the prompt-level shadow of H2.

4.3 Causal results

The observational signal is robust and prompt-level predictive. The remaining question is whether manipulating the signal at its strongest site recovers refusal behavior. Two complementary interventions are tested: (i) cross-condition patching of the refusal-direction component from a clean $k = 0$ source state, and (ii) direct additive injection of a scaled refusal-direction unit vector. Each is run at three layers and multiple early generated-token positions.

H4: cross-condition patching from a clean source. For each prompt, the model is run once under $k = 0$ to extract the refusal-direction component at the last-input-token position, and once under $k = 3$ to generate the attacked response. During the $k = 3$ run, the refusal-direction component at one early generated-token position is replaced with the clean component extracted from the $k = 0$ run on the same prompt. The grid is layers $\{16, 24, 27\} \times$ target positions $\{0, 1, 3, 5\} \times 25$ prompts = 300 records (Table 4, Figure 4). No cell has a restoration rate above 0.08; every 95% bootstrap CI overlaps zero; McNemar’s test cannot fire in any cell because the discordant count is below 5 everywhere. H4 is *not* supported under this intervention setting.

H5/H6: additive direction injection with controls. The additive intervention tests a tighter causal claim than patching: it adds $\alpha \cdot \hat{d}_\ell$ directly to the residual at one generated-token position. The grid is layers $\{16, 24, 27\} \times$ positions $\{0, 1, 3\} \times \alpha \in \{1.0, 2.0\} \times 25$ prompts $\times$ three direction types: the real refusal direction; a random unit vector (5 seeds); and a unit vector orthogonalized against the refusal direction (5 seeds). Total 4,950 attacked-condition generations. Multi-seed random and orthogonal control runs define the noise floor for “what does any single-position residual

Table 4: Cross-condition patching ($k = 0$ source $\rightarrow k = 3$ target). $n = 25$ prompts per cell; bootstrap 95% CIs over per-prompt outcomes ($B = 2000$). **Conclusion to draw:** no cell exceeds 0.08 restoration; every CI overlaps zero. H4 not supported.

Layer	tt	Restored	Lost	Restoration rate	95% CI
16	0	1	0	0.04	[0.00, 0.12]
16	1	0	2	0.00	[0.00, 0.00]
16	3	1	0	0.04	[0.00, 0.12]
16	5	0	0	0.00	[0.00, 0.00]
24	0	2	2	0.08	[0.00, 0.20]
24	1	0	4	0.00	[0.00, 0.00]
24	3	0	1	0.00	[0.00, 0.00]
24	5	1	0	0.04	[0.00, 0.12]
27	0	0	0	0.00	[0.00, 0.00]
27	1	1	2	0.04	[0.00, 0.12]
27	3	1	0	0.04	[0.00, 0.12]
27	5	1	0	0.04	[0.00, 0.12]

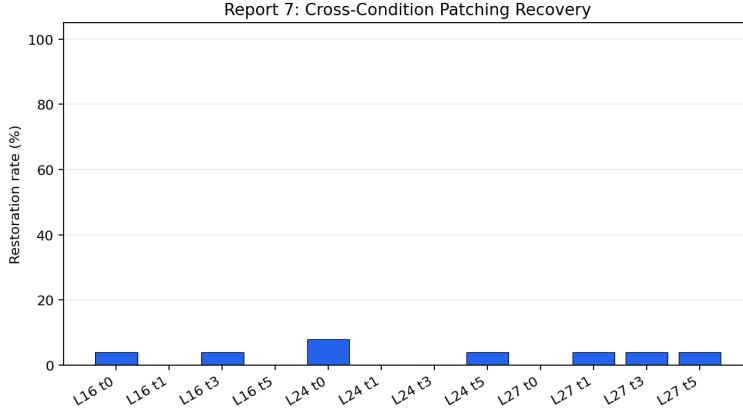


Figure 4: Cross-condition patching restoration rate by (layer, target position). **Conclusion to draw:** all cells ≤ 0.08 ; null is robust across layers and early generated positions.

edit do to behavior at this site?” Table 5 reports the mean restoration rate by direction type at the late layers, pooled across positions and α ; Table 6 reports the per-cell rates for the refusal direction.

Table 5: Mean restoration rate by direction type at late layers (24, 27), pooled across positions and α . Random and orthogonal are averaged over 5 seeds each. **Conclusion to draw:** the refusal direction restores refusal in zero of 12 late-layer cells (300/300 unchanged); random and orthogonal controls average $\approx 0.5\%$. The refusal direction is statistically indistinguishable from random noise: H5 strongly null, H6 consistent with that null.

Direction type	Mean restoration rate	Max	n cells
Refusal	0.000	0.00	12
Random	0.005	0.02	12
Orthogonal	0.004	0.01	12

Benign positive control (specificity test for H6). The same additive intervention is applied to 25 benign Alpaca prompts at $k = 0$, with the same grid (3 layers $\times$ 3 target positions $\times$ 2 α values $\times$ 3 direction types). The *intended* reading of “restoration” on benign prompts is false-refusal induction: if the refusal direction is causally active on the chat model independent of context, adding $\alpha \cdot \hat{d}_\ell$ to a benign residual should make the model refuse a benign request. We observe *zero* false-refusals across all 54 benign cells. Combined with the harmful null in Table 5, this indicates the intervention

Table 6: Per-cell additive intervention restoration rate at the late layers (refusal direction only); all 12 cells return zero. **Conclusion to draw:** even with the strongest available causal lever (the empirically extracted refusal direction, added at the cleanest restoration target position 0 with $\alpha = 2$), no cell flips behavior at the late layers.

Layer	tt	$\alpha = 1.0$	$\alpha = 2.0$
24	0	0.00	0.00
24	1	0.00	0.00
24	3	0.00	0.00
27	0	0.00	0.00
27	1	0.00	0.00
27	3	0.00	0.00

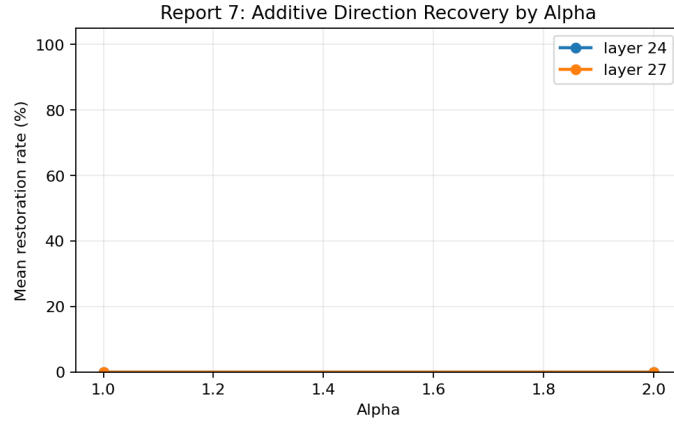


Figure 5: Additive direction restoration rate by α , late layers. **Conclusion to draw:** no α moves the refusal direction above the random/orthogonal noise floor at the late layers.

at this site is essentially causally inert in either direction, ruling out the simplest “the direction is a free knob” account of the prior observational results.

5 Conclusions

Status of each hypothesis. H1, H2, and H3 are supported; H4 and H5 are not supported in the direction they predict; H6 is supported in the negative direction (refusal vs. random/orthogonal controls are statistically indistinguishable, and the benign positive control produces no false-refusals). Table 7 summarizes.

Table 7: Hypothesis status.

H	Status	Note
H1	Supported	$0.92 \rightarrow 0.32$ at $k = 3$; 0.36 at $k = 10$.
H2	Supported (held-out)	Monotone deltas 0.76, 1.63, 3.23, 4.08 at L16/20/24/27.
H3	Supported	Refuser-complier gap +0.93, +1.39, +1.85 at L20/24/27.
H4	Not supported	All cells ≤ 0.08 restoration; CIs overlap 0.
H5	Not supported (strongly)	Refusal direction restores 0 of 300 prompts at L24/L27.
H6	Supported in the negative direction	Refusal vs. controls indistinguishable; benign positive control: 0 false-refusals.

Interpretation. The Report 5 design treated information loss and active suppression as two competing mechanistic accounts of the attack. The discriminator was: if a clean intervention restores refusal, the information-loss account is supported; if it does not, then the late-layer signal is more likely to be a downstream readout or to be overridden by other computations. Two cleaner interventions with controls both fail at the late layers, even though the observational signal at those same layers is unambiguous: the refusal direction is robust under a held-out extraction set (H2), tracks prompt-level behavior under attack (H3), and additive injection at early generated positions does nothing distinguishable from random or orthogonal controls (H5/H6).

The safest interpretation is therefore not simple information loss. The late-layer refusal-direction projection appears to be a real and predictive readout of the model’s attacked state, but under the intervention granularity tested here it is not a direct behavioral knob. One plausible explanation is active suppression or “locking-in”: by the time generation begins, the prefill has already shaped the model state enough that a single-position residual edit in layers 24/27 is too local and too late to recover refusal. This does not prove the full suppression mechanism, because multi-position edits, earlier residual interventions, or attention-routing interventions were not tested. It does, however, rule out the simpler account that restoring or adding the late-layer refusal-direction component at one generated-token position is sufficient to recover refusal. The benign positive control reinforces this by showing the direction does not act as a context-independent causal knob.

Threats to internal validity.

- *Classifier validity.* A stratified spot-check of 32 outputs was independently re-labelled by a secondary classifier (a strong LLM used as an independent rater); auto-vs-secondary agreement is 87.5% (28/32) overall and 100% on benign $k = 0$ and harmful $k = 0$. The 3 flips concentrate in attacked conditions: two cases where the model pivots to a safer related topic without using a refusal phrase (“soft refusal via redirect”) and one case where the disclaimer contained the word “illegal” and produced a false-positive refusal label. The reported attacked refusal counts may be off by ± 1 –2 prompts on each side; the $k = 0$ baseline and the benign control are very reliable.
- *Intervention scope.* Every interventional result is for a single one-time residual edit at one generated-token position. Multi-position edits, multi-token-window edits, earlier-layer edits, and attention-routing edits are not tested. The active-suppression reading is bounded by this scope.
- *Sample size.* $n = 25$ prompts per cell. Bootstrap CIs are wide and McNemar cannot fire in any patching cell. A larger sample could detect small effects below this resolution but would not change the qualitative finding given the magnitude of the controls’ noise floor (≤ 0.02).
- *Direction extraction.* The refusal direction is computed via difference-in-means on a held-out 50+50 prompt set disjoint from the 25 traced prompts. The direction is therefore not fit

on the prompts it is evaluated on, which addresses an overfitting concern raised in earlier work.

- *Layer-magnitude comparability.* Mean projections at different layers live on different scales, so cross-layer numeric comparisons describe direction and ordering, not raw magnitude.

Threats to external validity.

- *Single model.* Only Llama-3.1-8B-Instruct is tested. The results may not transfer to substantially larger or differently-aligned chat models, or to base models.
- *Single attack family.* Only prefilling is tested. The conclusion about late-layer signal causality may not generalize to suffix attacks [7] or other adversarial input perturbations.
- *Single classifier.* The phrase-list classifier is used as the behavioral label across all analyses. A different classifier family (e.g. trained safety classifier) might shift attacked refusal counts; the validity check addresses this for the spot-checked subset but not across the full distribution.

Future work. The clear next step is to test interventions at the sites this study does not reach: multi-position residual edits across the prefill window, attention-head and attention-routing interventions in late layers, and earlier-layer additive injections that act on the model state before generation begins. If any of these recover refusal, the active-suppression-or-locking-in reading becomes more concrete; if none do, the picture moves further toward distributed safety implementation along the lines suggested by Chen et al. [2] and Zhou et al. [6]. A parallel line is to replicate the observational story (H1–H3) on additional aligned chat models to test whether the late-layer monotone-by-depth signature is specific to Llama-3-family weights or is a general feature of shallow alignment.

Summary. This study contributes a clean and now-robust observational result — a monotone-by-depth late-layer negative shift of the refusal-direction projection under prefilling, that predicts behavior at the prompt level — and a clean negative causal result — two complementary single-position interventions on that signal fail to restore refusal, with multi-seed direction controls and a benign positive control showing the null is not a direction-quality artifact. The result narrows the mechanism of prefilling attacks on Llama-3.1-8B-Instruct from “the refusal-direction signal is removed and putting it back fixes things” to “the late-layer refusal-direction signal is a readout of an earlier or distributed decision that single-position residual interventions cannot overturn,” subject to explicit caveats about intervention scope.

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